

Guide for calculating the decomposition of changes in poverty according to Datt and Ravallion (1992)

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This text is a guide for calculating income poverty decomposition following the methodology proposed by Datt and Ravallion (1992). The first section provides an analytical description of each of the components or “effects” that are taken into account. The second part of the text is devoted to outlining, step by step, the workflow that must be followed to obtain this decomposition. The last section presents an application of the method for the case of Mexico between 2018 and 2022. The appendix presents the R script or routine containing the functions necessary to perform the calculations directly. The Github repository with the codes and databases is located at: https://github.com/rafaelpinedaa/D_R_Descomp.

1. Descomposition definition

The Datt and Ravallion (1992) decomposition is an approximation for determining how much of the absolute change in the prevalence of income poverty² is due to economic growth and how much to changes in income distribution. The growth effect (G) is captured through the change in average income, keeping the distribution constant (i.e., without changing the Lorenz curve), and the redistribution effect (RD) through changes in distribution (how the Lorenz curve shifts), keeping average income constant.

Changes in income and inequality interact with each other; therefore, the sum of the two individual effects does not exactly give the value of the absolute change in poverty. It partially creates a residual term (R) as a third component (Ravallion, 2016, p. 251). However, this residual, in addition to containing this interaction, also incorporates additional effects from other factors that affect change but are not

¹ The author thanks Dr. Enrique Minor Campa for his advice in preparing the calculations.

² From now on, the word “poverty” will be used, but at all times we will be referring to income poverty.

explicitly reflected in the two effects mentioned³. Later, we'll see how to incorporate these types of factors that aren't considered in the original decomposition.

If we represent the poverty measure in period t as P_t , then:

$$P_t = P(z | Y_t, L_t)$$

Where poverty is a function of the poverty line (z), given the mean income (Y_t) and the Lorenz curve (L_t), both evaluated at time t . Note that in the original equation the poverty line does not have a temporal subscript t , indicating that the decomposition assumes it to be a constant over time. For ease of notation in what follows⁴, this constant is defined as equal to 1.

1.1 Original decomposition (Datt and Ravallion, 1992)

Let P_1 and P_2 represent poverty data at two different points in time (where $t=1$ is the initial year and $t=2$ is the final year). The change in poverty prevalence can then be decomposed as follows:

$$P_2 - P_1 = \Delta P$$

$$\Delta P = P(Y_2, L_2) - P(Y_1, L_1) = G + RD + R$$

Where G is the growth component, RD is the redistribution component, and R represents the residual or interaction effect. These components are defined as follows:

$$G = P(Y_2, L_1) - P(Y_1, L_1)$$

$$RD = P(Y_1, L_2) - P(Y_1, L_1)$$

$$R = \Delta P - G - RD$$

³ The size of the residual is one of the main criticisms of the original decomposition, along with the indicator's lack of asymmetry (which reflects its sensitivity to the choice of base year). Therefore, this decomposition can be interpreted only as a first approximation, not a definitive one. See Millán (2014) for a proposed method that attempts to correct these shortcomings.

⁴ This assumption is not insignificant, as it implies that income must be deflated relative to a base period to compare poverty measures over time. The following section explores this topic in more detail.

Where the G effect is interpreted as the difference between the poverty measure of the initial year maintaining that same distribution, but with the average income of the final year $[P(Y_2, L_1)]$; and between the observed data of the initial year⁵ $[P(Y_1, L_1)]$. The redistribution effect is the difference between the poverty of the initial year with the same average income of that year, but with the distribution of the final year $[P(Y_1, L_2)]$; and between the observed data of the initial year⁶. The residual is the difference between what poverty actually changed (ΔP) and the two previous effects.

1.2 Inflation effect

If we remove the assumption that the poverty line z is constant, it is possible to recover part of the residual term⁷, that is, the effect of the change in the monetary value of the lines. In other words, it allows us to understand the magnitude of the effect of inflation on changes in poverty. For Mexico, the monetary value of the poverty line is updated each month based on movements in the National Consumer Price Index for the corresponding items. To capture the effect of changes in the value of the line, the average income (but projected to pesos for the final year) and the distribution of the initial period are kept fixed, and the poverty line for the final period is used, then the difference is calculated with respect to the observed data for the initial period. That is, the inflation effect Π will be equal to:

$$\Pi = P(z_2|Y_{1,proy}, L_1) - P(z_1|Y_1, L_1)$$

Where the effect Π is the difference between the poverty measure calculated using the poverty line for the final year, given the average income of the initial year but projected to final pesos and the same distribution; and the original measure for the initial year.

⁵ Intuitively, the question that this effect answers is: "What would be the poverty level in a subsequent year if the income level changed in the way it did in reality, but if the income distribution had remained constant between this point in time and the previous one used as a reference?" (Cortés et al., 2003, p. 310)

⁶ This effect answers the question: "What would be the poverty level in a given year if the population's income had not changed during the period under analysis, but the income distribution had recorded the changes observed in reality?" (Cortés et al., 2003, p. 310).

⁷ Remember that the residual incorporates everything we don't explicitly specify in the decomposition. Therefore, in addition to the interaction between the growth effect and redistribution, it also includes other components that explain changes in the prevalence of poverty; for example, the effect of inflation (Minor, 2014, p. 32).

1.3 Effect by some source of income

According to Minor (2014, p. 33), retrieved in Cortés (2014), it is possible to isolate the effect of government cash transfers on the prevalence of poverty. This effect is captured by calculating the differences in poverty data with and without social transfers for each year, and then the change between those two differences. This can be generalized to any income source k , based on the same logic as above:

$$K = [P(z_2|Y_2, L_2) - P(z_2|Y_{2,SK}, L_{2,SK})] - [P(z_1|Y_1, L_1) - P(z_1|Y_{1,SK}, L_{1,SK})]$$

Where K is the effect of the given source, and the subscripts SK represent measures that do not consider this income source (i.e., by removing them from the calculations). It's important to note that to perform this decomposition, the effect must first be calculated; subsequently, to calculate the other effects, data that don't include the respective source of income are used to isolate it.

2.1 Instructions for calculating the original decomposition

The final poverty databases published by CONEVAL must be used.

1. Official poverty data are obtained for periods 1 and 2, and their change is calculated. This change is the target data. Indeed, in the end, the sum of the effects (including the residual) must result in the absolute change in poverty.
2. Se deflacta la base del año 2 a pesos de agosto del año 1.
3. The base for year 2 is deflated to pesos as of August of year 1.
4. The average total current income per capita (ICTPC) is calculated for each year. **In the case of year 2, the calculation corresponds to the already deflated ICTPC.**
5. The growth rate of the ICTPC between period 1 and the deflated ICTPC for period 2 is obtained (g_G).

6. Using the data for year 1, g_G is applied to all ICTPC observations⁸. This carries over the income from period 1 to period 2, keeping the distribution fixed.
7. Poverty is estimated using these new incomes. Next, we calculate the difference with respect to the values observed in year 1:

$$G = P(Y_2^*, L_1) - P(Y_1, L_1)$$

It holds that $Y_2^* = Y_1(1 + g_G)$

8. For the redistribution effect, we use the inverse of g_G , so $g_{RD} = \frac{1}{g_G}$. Esto se This is because we want to know the change in the average ICTPC, now using period 2 as a reference.
9. Using base 2, we apply the g_{RD} rate to all ICTPCs deflated to pesos from August of year 1. This means that we carry the distribution from period 2 to period 1, maintaining the same average ICTPC (Recommendation: check at this point that the average income data is the same as the one previously calculated using base 1).
10. Poverty is estimated using the same line as in period 1 (not the lines from period 2) and the difference is calculated with respect to the data observed in period 1:

$$RD = P(Y_1^*, L_2) - P(Y_1, L_1)$$

The following holds: $Y_1^* = Y_2(1 + g_{RD})$.

11. The two effects obtained, $G+RD$, are added together and subtracted from the absolute change in poverty to obtain the value of the residual effect:

$$R = \Delta P - (G + RD)$$

Therefore, the following always holds: $\Delta P = G + RD + R$

⁸ Important to remember: Growth rate = $g_G = \left(\frac{ICTPC_{2,deflactado}}{ICTPC_1} - 1 \right)$. To apply the growth rate to all income $1 + g_G$ is used, that is: $\left(\frac{ICTPC_{2,deflactado}}{ICTPC_1} \right)$

2.2 Instructions for calculating the inflation effect

To calculate the inflation effect, it is assumed that the original decomposition has been performed. The goal is to reduce the value of the residual, controlling for the effect of the change in the price level. Therefore, the two initial effects are not altered.

1. We project the ICTPC for year 1 to the pesos of August of year 2. To do this, we can use the same deflator as above, but instead of dividing income by that value, we multiply it.
2. Poverty is estimated taking into account the value of the lines corresponding to period 2.
3. The difference between the data obtained and the observed figures for period 1 is obtained to obtain the inflation effect.

$$\Pi = P(z_2|Y_{1,proy}, L_1) - P(z_1|Y_1, L_1)$$

4. The three effects are added together and the difference is calculated with respect to the absolute change in prevalence to obtain the new value of the residual:

$$R = \Delta P - (G + RD + \Pi)$$

2.3 Instructions for calculating the effect of some source of income

Following Minor's (2014) proposal, for illustrative purposes, the effect of social program transfers is considered as a source of income.

1. We begin by calculating the effect of this source on the change in poverty. To do this, poverty estimates are made with and without this source for each year, considering the original bases (i.e., without deflation and using the respective poverty lines for each year). The effect of social programs will then be:

$$K = (CPS_2 - SPS_2) - (CPS_1 - SPS_1)$$

Where CPS and SPS refer to the estimated poverty data with and without social programs within the ICTPC, respectively, for year 2 and year 1.

2. Now, the generated bases will be used, where the source of income from social program transfers is removed. That is, base 1 and base 2 will now be base 1 without government transfers and base 2 without government transfers. From there, the exact same steps as the original decomposition and the inflation effect are followed, taking into account the bases on which the calculations will be made. Now, it turns out that:

$$P(Y_1, L_1) = P(Y_{1,SPS}, L_{1,SPS})$$

$$P(Y_2, L_2) = P(Y_{2,SPS}, L_{2,SPS})$$

3. Once the complete decomposition into growth, redistribution, inflation, and the corresponding source is achieved, the residual is calculated using the same logic as before:

$$R = \Delta P - (G + RD + \Pi + K)$$

Important: The interpretation of the growth, redistribution, and inflation effects changes when an income source is isolated. The three effects now reflect their respective interpretations, but on total current per capita income, which does not take into account the selected income source. For example, in the case of social transfers, the growth effect only captures the rate of income growth, excluding the growth in social program transfers; the same applies to the remaining two effects. It is important to take this detail into account when presenting the results.

3. The decomposition in the case of Mexico

There are at least four previous works that analyze changes in poverty in Mexico based on the decomposition proposed by Datt and Ravallion⁹. Cortés et al. (2002) highlight that, throughout the 1990s, the change in the prevalence of poverty is explained primarily by economic growth and not so much by changes in income distribution. That is, the change in poverty is substantially due to changes in the average income of the population; this result contrasts with that found by Székely

⁹ See Székely (1998), Cortés *et al.* (2002), Cortés (2014) and Ochoa (2019).

(1998) for the period 1984 to 1992, where persistent increases in poverty are attributed to redistribution and, to a lesser extent, to economic growth.

The two previous texts present estimates of the original decomposition. In Cortés (2014), the analysis is extended to take into account inflation components and the effect of federal government social programs. By applying the methodology to pairs of years starting in 2000, the author finds that the effects of social programs have tended to counteract the negative effects of rising prices and unfavorable economic growth, thus concluding that "poverty would have increased much more if social programs had not acted as palliatives" (Cortés, 2014, p. 27).

Furthermore, the author highlights that the magnitude of the effect of federal government transfers is greater on food poverty than on capabilities and wealth poverty, which could lead to the conclusion that these programs were relatively well targeted at the lowest-income population.

Finally, a recent study by Ochoa (2019) shows that, when adjusting ENIGH data to national accounts, the poverty decomposition yields results opposite to those obtained without adjustment. In this case, it is shown that economic growth, despite being low, contributed more to poverty reduction than the effect of income distribution for the period 2014-2016.

3.2 Application of the decomposition for Mexico, 2018-2022

This section presents an application of the material presented in the previous sections, following the same order in which the different types of decompositions were presented. The analysis period is 2018-2022; the year 2020 is omitted, taking into account the drastic change in average income due to the health crisis caused by the economic disruption caused by COVID-19. An analysis of the decomposition is also included, differentiating between four major regions.

Between 2018 and 2022, income poverty fell by 6.4 percentage points (pp), from 49.9% to 43.5% of the Mexican population; for extreme income poverty, the reduction was much more modest at 1.9 pp, from 14% to 12.1%, respectively. Total

current income per capita increased by 7.8% in real terms, from 4,516 in 2018 to 4,873 in 2022, in 2018 pesos.

Table 1 presents the results of the classic Datt and Ravallion decomposition. It is notable that the dominant effect on poverty reduction is economic growth, as it explains -4.5 pp of the -6.4 pp that represent the total change in prevalence; the redistribution effect also contributes to this result, although to a lesser extent, with an incidence of -3.0 pp. On the other hand, the opposite is true for extreme poverty: here the effect of changes in income redistribution dominates the growth effect; this sheds light on the fact that the benefits of economic growth do not necessarily reach the pockets of the poorest population.

However, the astute reader will have noticed that the residuals obtained in both cases are relatively high, especially for extreme poverty. Although this is a good first snapshot, the level of these residuals calls for extending the analysis to take into account other additional effects not captured by this decomposition.

Table 1. Original decomposition of changes in poverty for Mexico, 2018-2022.

	Observed data			Effects		
	2018	2022	Δ Poberty	Growth (G)	Redistribution (RD)	Residual (R)
Income poverty	49.9	43.5	-6.4	-4.5	-3.0	1.1
Extreme income poverty	14.0	12.1	-1.9	-2.1	-2.5	2.7

Fuente: Elaboración propias con datos de la ENIGH 2018 y 2022.

By abandoning the assumption that poverty lines are the same from one year to the next, we can capture the effect of price level changes on poverty, reflected in the change in the monetary value of the respective baskets. In 2018, the value of the food and non-food baskets (i.e., the income poverty line) was \$3,325 for urban areas and \$2,317 for rural areas; by 2022, these values were \$4,158 and \$2,971 pesos, respectively. By 2018, the value of the urban food basket (the extreme income poverty line) was \$1,544 pesos, and the rural food basket was \$1,165 pesos; while by 2022, they were \$2,086 and \$1,600 pesos, respectively. If we take the simple

average of the rural-urban lines and calculate the growth rates, we find that the income poverty line increased by 26.4% between 2018 and 2022, while the extreme poverty line increased by 36.1%. This difference makes sense when considering that the extreme poverty line is made up entirely of food, and this is where one of the greatest sources of inflationary pressure has been found.

Table 2 adds the inflation effect to the previous breakdown, showing that neither the growth nor the redistribution effect were affected, while the residual effect was practically reduced to zero. In the case of income poverty, the impact of the increase in the price level exerts a 1.2 pp pressure on the prevalence of poverty, which is easily offset by either the economic growth component or the redistribution component.

Table 2. Inflation-related decomposition of changes in poverty for Mexico, 2018-2022.

	Datos observados			Efectos			
	2018	2022	Δ Pobreza	Crecimiento (G)	Redistribución (RD)	Inflación (π)	Residual (R)
Pobreza por ingresos	49.9	43.5	-6.4	-4.5	-3.0	1.2	-0.1
Pobreza extrema por ingresos	14.0	12.1	-1.9	-2.1	-2.5	2.9	-0.3

Fuente: Elaboración propias con datos de la ENIGH 2018 y 2022.

However, the story is again different for extreme poverty by income. In this case, the effect of the increase in food prices pushes the prevalence of extreme poverty upward by 2.9 percentage points (i.e., more than double that of income poverty). This is important since the magnitude of this component is even greater than the growth effect and the redistribution effect. In other words, the small gains obtained from economic growth or from changes in income distribution among the lowest-income population are completely offset by inflation.

Now, to understand the effect of a specific source of income, we will use income derived from social programs (SP). Table 3 presents, in the first part, a static analysis of the impact of SP on the level of poverty and extreme poverty by income.

The "instantaneous" effect of SP in 2018 was -1.4 for poverty and -2.5 for extreme poverty; That is, in the hypothetical case that these benefits were suddenly discontinued, poverty would increase by 1.4 percentage points and 2.5 percentage points, respectively, slightly more so for the extremely poor. In 2022, with the removal of the PS, poverty would increase by 2.9 percentage points and extreme poverty by 3.2 percentage points.

Table 3. Effect of social programs on changes in poverty in Mexico, 2018-2022.

	2022			2018			Diferencia de la diferencia (G = C - F)
	CPS (A)	SPS (B)	Diferencia (C = A - B)	CPS (D)	SPS (E)	Diferencia (F = D - E)	
Pobreza por ingresos	43.5	46.4	-2.9	49.9	51.3	-1.4	-1.5
Pobreza extrema por ingresos	12.1	15.2	-3.2	14.0	16.5	-2.5	-0.7

Fuente: Elaboración propias con datos de la ENIGH 2018 y 2022.

Nota: CPS (Con Programas Sociales), SPS (Sin Programas Sociales)

We need to differentiate the two years for each type of poverty to understand the effect over the period. In doing so, we find that the effect of social programs on the change in poverty is -1.5 for poverty and -0.7 for extreme poverty. This means that, if we consider the entire period and not each year in isolation, social programs, despite contributing to poverty reduction, are doing so more for the less poor; since the effect on extreme poverty is half of the effect on income poverty.

By reviewing the full picture, we can analyze the new values for the growth effect and redistribution that do not take into account income derived from social programs and their relationship with the inflation effect. For income poverty, the economic growth component remains the main explanatory factor for poverty reduction: of the -6.4 percentage points of the change in prevalence, -3.5 percentage points are associated with the change in the average income of the population without HC. Changes in income distribution contribute -2.5% to this decline, while the inflationary effect pushes it upward by 1.1 pp. The impact of social programs, as mentioned, is -

1.5%, thus fully covering, and slightly more than, the negative effect of price increases.

Tabla 4. Decomposition of the effect of social programs on changes in poverty in Mexico, 2018-2022.

	Datos observados			Efectos				Residual (R)
	2018	2022	Δ Pobreza	Crecimiento (G)	Redistribución (RD)	Inflación (π)	Programas sociales (K)	
Pobreza por ingreso	49.9	43.5	-6.4	-3.5	-2.5	1.1	-1.5	0.0
Pobreza extrema por ingreso	14	12.1	-1.9	-1.6	-2.2	3.1	-0.7	-0.5

Fuente: Elaboración propias con datos de la ENIGH 2018 y 2022.

Nota: CPS (Con Programas Sociales), SPS (Sin Programas Sociales)

In the case of extreme poverty by income, the story is similar to that found with the two previous decompositions: the inflationary effect superimposes the effect of growth, redistribution, and social programs. Of the -1.9 percentage points reduced in prevalence, only -1.6 are explained by economic growth, slightly more by changes in income distribution (-2.2), and only -0.7 by social programs; at this point, not even the sum of the growth effect and the social programs alone is sufficient to provide the necessary safety valve to address the pressure generated by inflation.

Finally, following the regionalization proposed by Bassols (1979), it is possible to analyze the determinants of changes in the prevalence of poverty for four regions of the country. The complete results are found in Table 5.

Tabla 5. Poverty decomposition by region, Mexico, 2018-2022..

		Datos observados			Efectos			Residual (R)
		2018	2022	Δ Pobreza	Crecimiento (G)	Redistribución (RD)	Inflación (π)	
Septentrional	Pobreza por ingresos	39.9	30.6	-9.3	-7.4	-3.8	1.1	0.8

	Pobreza extrema por ingresos	8.0	6.8	-1.2	-2.2	-0.7	2.2	-0.5
Central	Pobreza por ingresos	48.0	44.0	-3.9	-1.2	-4.0	1.4	-0.1
	Pobreza extrema por ingresos	10.4	10.2	-0.1	-0.6	-2.2	2.8	-0.1
Sur	Pobreza por ingresos	74.2	65.7	-8.6	-8.8	-0.6	0.8	0.0
	Pobreza extrema por ingresos	39.9	30.4	-9.5	-9.1	-4.5	4.5	-0.4
Oriental	Pobreza por ingresos	59.2	51.3	-7.9	-6.8	-1.3	1.1	-0.9
	Pobreza extrema por ingresos	20.2	15.5	-4.8	-4.1	-3.7	3.9	-0.9

Fuente: Elaboración propias con datos de la ENIGH 2018 y 2022.

As a starting point, it is important to note that the Southern region saw the largest decrease in extreme income poverty of all the regions considered, falling from 39.9% in 2018 to 30.4% in 2022 (i.e., a decrease of -9.5 percentage points). This was followed by the Eastern region with a decrease of -4.8 percentage points, then the Northern region with a much more modest drop of -1.2 percentage points, and finally the Central region with virtually the same levels observed year-over-year. Regarding income poverty, the largest decline was recorded in the Northern region with -9.3 pp, followed by the Southern, Eastern, and Central regions with decreases of -8.6, -7.9, and -3.9 pp, respectively, between 2018 and 2022.

Regarding the analysis of each of the effects to explain the dynamics of extreme poverty, the main findings can be summarized as follows:

- The growth effect was much more pronounced in the Southern region: its contribution to the reduction of extreme poverty was -9.1 pp. That is, the change in average income had a greater impact on reducing poverty in this

region compared to the rest. On the other hand, the Central region had the smallest impact from this effect, with only -0.6 pp.

- In the case of the redistribution effect, the picture is similar to the previous case: the Southern region has the greatest impact via this effect, equivalent to -4.5 pp. While the Northern region shows a much smaller redistribution effect, of only -0.7 pp.
- The inflation effect also has its greatest impact on the Southern region, as it increased extreme poverty by 4.5 pp. This denotes the markedly regressive nature of the price increase, highlighting how the most disadvantaged are most sensitive and vulnerable to these types of shocks. On the other hand, in the Northern region, this effect had an impact of 2.2 pp.

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Appendix. R functions for calculating the Datt and Ravallion (1992) decomposition.

Three R functions are available for users to perform the calculations directly. Each function corresponds to one of the decomposition types reviewed in this guide: the original decomposition, the inflation-adjusted decomposition, and the income source decomposition.

The arguments to be taken into account for the original decomposition function (DR_descomp.R) are:

- A: The database containing the information for the base or reference year.
- B: The database containing the information for the final or comparison year.
- def: This is the deflator corresponding to year B in pesos from year A.
- lp1_urb: Urban extreme poverty line by income for year A.
- lp1_rur: Rural extreme poverty line by income for year A.
- lp2_urb: Urban poverty line by income for year A.
- lp2_rur: Rural poverty line by income for year A.

```

1. #####
2. ##### Descomposición original #####
3. #####
4.
5. DR_descomp <- function(A, B, def, lp1_urb, lp1_rur, lp2_urb, lp2_rur) {
6.
7.   library(tidyverse)
8.   library(readxl)
9.   library(haven)
10.  library(survey)
11.  library(srvyr)
12.
13.  # FUNCIÓN PARA ESTIMAR POBREZA POR INGRESOS
14.  pobrezaing <- function(base) {
15.    baseX <- mutate(base,
16.      # Se identifica a los hogares bajo lp1
17.      plp_e=case_when(ictpc <lp1_urb & rururb==0 ~ 1,
18.        ictpc>=lp1_urb & rururb==0 & !is.na(ictpc) ~ 0,
19.        ictpc <lp1_rur & rururb==1 ~ 1,
20.        ictpc>=lp1_rur & rururb==1 & !is.na(ictpc) ~ 0),
21.      # Se identifica a los hogares bajo lp2
22.      plp =case_when(ictpc <lp2_urb & rururb==0 ~ 1,
23.        ictpc>=lp2_urb & rururb==0 & !is.na(ictpc) ~ 0,
24.        ictpc <lp2_rur & rururb==1 ~ 1,
25.        ictpc>=lp2_rur & rururb==1 & !is.na(ictpc) ~ 0))
26.  }
27.
28.  # Fijamos los objetos de diseño muestral para las bases originales
29.  svyA <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = A)
30.  svyB <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = B)
31.
32.  # Dato observado para A (año inicial/de referencia) y para B (año final/de
    comparación)
33.  ObsA <- as.numeric(svymean(~plp+plp_e, design = svyA))
34.  ObsB <- as.numeric(svymean(~plp+plp_e, design = svyB))
35.
36.  # Obtenemos el cambio absoluto en pobreza
37.  dif <- (ObsB - ObsA)
38.
39.  ##### EFECTO CRECIMIENTO #####
40.  # Deflactamos los ingresos de B a pesos de A
41.  baseB_EC <- mutate(B, ictpc=ictpc/def)
42.
43.  # Nuevo objeto de diseño
44.  svyB_EC <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = baseB_EC)
45.
46.  # Tasa de crecimiento real del ictpc entre A y B
47.  rate_A_B <- as.numeric(svymean(~ictpc, design = svyB_EC)/svymean(~ictpc,
    design = svyA))
48.  rate_B_A <- as.numeric(svymean(~ictpc, design = svyA)/svymean(~ictpc,
    design = svyB_EC))
49.
50.  # Aplicamos la tasa de crecimiento real del ictpc a los ingresos de A
51.  baseA_EC <- mutate(A, ictpc = ictpc*rate_A_B)
52.
53.  # Calculamos pobreza con líneas de A
54.  baseEC <- pobrezaing(baseA_EC)
55.
56.  # Nuevo objeto de diseño
57.  svyEC <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = baseEC)
58.
59.  EC <- as.numeric(svymean(~plp+plp_e, design = svyEC))
60.  G <- (EC - ObsA)
61.
62.  ##### EFECTO REDISTRIBUCIÓN #####
63.  # Deflactamos a pesos de A y aplicamos la tasa de crecimiento de B/A
64.  baseB_ERD <- mutate(B, ictpc = (ictpc/def)*rate_B_A)

```



```

65.
66. baseB_ERD <- pobrezaing(baseB_ERD)
67.
68. svy_ERD <- svydesign(ids = ~upm , weights = ~factor,
  strata=~est_dis, data = baseB_ERD)
69.
70. ERD <- as.numeric(svymean(~plp+plp_e, design = svy_ERD))
71. RD <-(ERD - ObsA)
72.
73. # Generamos la de resultados y el residual
74. results <- round(data.frame("A" = ObsA, "B" = ObsB, "Cambio" = dif, "Crecimiento" = G,
  "Redistribución" = RD)*100,1)
75. results <- mutate(results, Residual = Cambio - (Crecimiento + Redistribución))
76. rownames(results) <- c("Pobreza por ingresos", "Pobreza extrema por ingresos")
77. results
78. }

```

Ejemplo de la salida en la consola de R:

```

> source("DR_descomp_Original.R")
>
> DR_descomp(baseA, baseB, def = 1.231968714, lp1_urb = 1544.07, lp1_rur = 1164.75,
+   lp2_urb = 3325.40, lp2_rur = 2316.57)

```

	A	B	Cambio	Crecimiento	Redistribución	Residual
Pobreza por ingresos	49.9	43.5	-6.4	-4.5	-3.0	1.1
Pobreza extrema por ingresos	14.0	12.1	-1.9	-2.1	-2.5	2.7

Para la descomposición tomando en cuenta el efecto inflación (DR_descomp_Inflación.R), es necesario agregar cuatro argumentos adicionales a los que se toman en cuenta en la descomposición original; consisten basicamente en las cuatro tipos de líneas de pobreza para el año B (final/de comparación):

- lp1_urb: Línea de pobreza extrema por ingresos urbana del año B.
- lp1_rur: Línea de pobreza extrema por ingresos rural del año B.
- lp2_urb: Línea de pobreza por ingresos urbana del año B.
- lp2_rur: Línea de pobreza por ingresos rural del año B.

```

1. #####
2. ##### EFECTO INFLACIÓN #####
3. #####
4.
5. DR_descompI <- function(A, B, def,lp1_urb, lp1_rur, lp2_urb,lp2_rur,Ilp1_urb, Ilp1_rur,
  Ilp2_urb,Ilp2_rur) {
6.
7.   library(tidyverse)
8.   library(haven)
9.   library(survey)
10.
11.  # FUNCIÓN PARA ESTIMAR POBREZA POR INGRESOS
12.  pobrezaing <- function(base) {
13.    baseX <- mutate(base,
14.      # Se identifica a los hogares bajo lp1
15.      plp_e=case_when(ictpc <lp1_urb & rururb==0 ~ 1,
16.        ictpc>=lp1_urb & rururb==0 & !is.na(ictpc) ~ 0,
17.        ictpc <lp1_rur & rururb==1 ~ 1,

```

```

18.             ictpc>=lp1_rur & rururb==1 & !is.na(ictpc) ~ 0),
19. # Se identifica a los hogares bajo lp2
20.     plp =case_when(ictpc <lp2_urb & rururb==0 ~ 1,
21.                   ictpc>=lp2_urb & rururb==0 & !is.na(ictpc) ~ 0,
22.                   ictpc <lp2_rur & rururb==1 ~ 1,
23.                   ictpc>=lp2_rur & rururb==1 & !is.na(ictpc) ~ 0))
24. }
25.
26. # Fijamos los objetos de diseño muestral para las bases originales
27. svyA <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = A)
28. svyB <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = B)
29.
30. # Dato observado para A (año inicial/de referencia) y para el año B (final/ de
    comparación)
31. ObsA <- as.numeric(svymean(~plp+plp_e, design = svyA))
32. ObsB <- as.numeric(svymean(~plp+plp_e, design = svyB))
33.
34. # Obtenemos el cambio absoluto en pobreza
35. dif <- (ObsB - ObsA)
36.
37. ##### EFECTO CRECIMIENTO #####
38. # Deflactamos los ingresos de B a pesos de A
39. baseB_EC <- mutate(B, ictpc=ictpc/def)
40.
41. # Nuevo objeto de diseño
42. svyB_EC <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = baseB_EC)
43.
44. # Tasa de crecimiento real del ictpc entre A y B
45. rate_A_B <- as.numeric(svymean(~ictpc, design = svyB_EC)/svymean(~ictpc,
    design = svyA))
46. rate_B_A <- as.numeric(svymean(~ictpc, design = svyA)/svymean(~ictpc,
    design = svyB_EC))
47.
48. # Aplicamos la tasa de crecimiento real del ictpc a los ingresos de A
49. baseA_EC <- mutate(A, ictpc = ictpc*rate_A_B)
50.
51. # Calculamos pobreza con líneas de A
52. baseEC <- pobrezaing(baseA_EC)
53.
54. # Nuevo objeto de diseño
55. svyEC <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = baseEC)
56.
57. EC <- as.numeric(svymean(~plp+plp_e, design = svyEC))
58. G <- (EC - ObsA)
59.
60. ##### EFECTO REDISTRIBUCIÓN #####
61. # Deflactamos a pesos de A y aplicamos la tasa de crecimiento de B/A
62. baseB_ERD <- mutate(B, ictpc = (ictpc/def)*rate_B_A)
63.
64. baseB_ERD <- pobrezaing(baseB_ERD)
65.
66. svy_ERD <- svydesign(ids = ~upm , weights = ~factor,
    strata=~est_dis, data = baseB_ERD)
67.
68. ERD <- as.numeric(svymean(~plp+plp_e, design = svy_ERD))
69. RD <- (ERD - ObsA)
70.
71. ##### EFECTO INFLACIÓN #####
72. baseA_EI <- mutate(A, ictpc = ictpc*def)
73.
74. baseA_EI <- mutate(baseA_EI,
75.                   # Se identifica a los hogares bajo lp1
76.                   plp_e=case_when(ictpc <lp1_urb & rururb==0 ~ 1,
77.                                   ictpc>=lp1_urb & rururb==0 & !is.na(ictpc) ~ 0,
78.                                   ictpc <lp1_rur & rururb==1 ~ 1,
79.                                   ictpc>=lp1_rur & rururb==1 & !is.na(ictpc) ~ 0),
80.                   # Se identifica a los hogares bajo lp2

```

```

81.         plp =case_when(ictpc <Ilp2_urb & rururb==0 ~ 1,
82.         ictpc>=Ilp2_urb & rururb==0 & !is.na(ictpc) ~ 0,
83.         ictpc <Ilp2_rur & rururb==1 ~ 1,
84.         ictpc>=Ilp2_rur & rururb==1 & !is.na(ictpc) ~ 0))
85.
86.   svy_EI <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = baseA_EI)
87.
88.   EI <- as.numeric(svymean(~plp+plp_e, design = svy_EI))
89.   I <- (EI - ObsA)
90.
91.   # Generamos la de resultados y el residual
92.   results <- round(data.frame("A" = ObsA, "B" = ObsB, "Cambio" = dif, "Crecimiento" = G,
93.   "Redistribución" = RD, "Inflación" = I)*100,1)
94.   results <- mutate(results,
95.   Residual = Cambio - (Crecimiento + Redistribución + Inflación))
96.   rownames(results) <- c("Pobreza por ingresos", "Pobreza extrema por ingresos")
97. }

```

Ejemplo de la salida en la consola de R:

```

> source("DR_descomp_Inflación.R")
> DR_descompI(baseA, baseB, def = 1.231968714, lp1_urb = 1544.07, lp1_rur = 1164.75,
+   lp2_urb = 3325.40, lp2_rur = 2316.57, Ilp1_urb = 2086.21, Ilp1_rur = 1600.18,
+   Ilp2_urb = 4158.35, Ilp2_rur = 2970.76)

```

	A	B	Cambio	Crecimiento	Redistribución	Inflación	Residual
Pobreza por ingresos	49.9	43.5	-6.4	-4.5	-3.0	1.2	-0.1
Pobreza extrema por ingresos	14.0	12.1	-1.9	-2.1	-2.5	2.9	-0.2

Finalmente, para el último tipo de descomposición, la de alguna fuente de ingreso (DR_descompSK.R), se necesitará un par de argumentos adicionales a los que se toman en cuenta cuando se obtiene el efecto inflación: las bases de datos que no tengan ingresos derivados de la fuente seleccionada (para fines del ejemplo de esta guía, los programas sociales).

- A_SK: La base de datos del año base o de referencia sin ingresos de la fuente elegida.
- B_SK: La base de datos del año final o de comparación sin ingresos de la fuente elegida.

```

1. #####
2. ##### FUENTE DE INGRESO #####
3. #####
4.
5. DR_descompSK <- function(A, B, A_SK, B_SK, def, lp1_urb, lp1_rur, lp2_urb, lp2_rur,
6.   Ilp1_urb, Ilp1_rur, Ilp2_urb, Ilp2_rur) {
7.   library(tidyverse)
8.   library(haven)
9.   library(survey)
10.
11.   # FUNCIÓN PARA ESTIMAR POBREZA POR INGRESOS
12.   pobrezaing <- function(base) {
13.     baseX <- mutate(base,
14.       # Se identifica a los hogares bajo lp1

```

```

15.         plp_e=case_when(ictpc <lp1_urb & rururb==0 ~ 1,
16.                         ictpc>=lp1_urb & rururb==0 & !is.na(ictpc) ~ 0,
17.                         ictpc <lp1_rur & rururb==1 ~ 1,
18.                         ictpc>=lp1_rur & rururb==1 & !is.na(ictpc) ~ 0),
19.         # Se identifica a los hogares bajo lp2
20.         plp =case_when(ictpc <lp2_urb & rururb==0 ~ 1,
21.                         ictpc>=lp2_urb & rururb==0 & !is.na(ictpc) ~ 0,
22.                         ictpc <lp2_rur & rururb==1 ~ 1,
23.                         ictpc>=lp2_rur & rururb==1 & !is.na(ictpc) ~ 0))
24.     }
25.
26.     # Preparamos los objetos de diseño de muestra
27.     svyA <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = A)
28.     svyB <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = B)
29.     svyA_SPS <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = A_SK)
30.     svyB_SPS <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = B_SK)
31.
32.     # Datos observados con y sin programas sociales
33.     ObsA <- as.numeric(svymean(~plp+plp_e, design = svyA))
34.     ObsB <- as.numeric(svymean(~plp+plp_e, design = svyB))
35.     ObsA_SPS <- as.numeric(svymean(~plp+plp_e, design = svyA_SPS))
36.     ObsB_SPS <- as.numeric(svymean(~plp+plp_e, design = svyB_SPS))
37.
38.     # Diferencia entre datos con y sin programas sociales
39.     Difa <- (ObsA - ObsA_SPS)
40.     DifB <- (ObsB - ObsB_SPS)
41.
42.     # Diferencia de la diferencia a través de los años
43.     Dif_PS <- DifB - Difa
44.
45.     # Obtenemos el cambio absoluto en pobreza
46.     dif <- (ObsB - ObsA)
47.
48.     ##### EFECTO CRECIMIENTO #####
49.     # Deflactamos los ingresos de B a pesos de A
50.     baseB_EC <- mutate(B_SK, ictpc=ictpc/def)
51.
52.     # Nuevo objeto de diseño
53.     svyB_EC <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = baseB_EC)
54.
55.     # Tasa de crecimiento real del ictpc entre A y B
56.     rate_A_B <- as.numeric(svymean(~ictpc, design = svyB_EC)/svymean(~ictpc,
design = svyA_SPS))
57.     rate_B_A <- as.numeric(svymean(~ictpc, design = svyA_SPS)/svymean(~ictpc,
design = svyB_EC))
58.
59.     # Aplicamos la tasa de crecimiento real del ictpc al los ingresos de A
60.     baseA_EC <- mutate(A_SK, ictpc = ictpc*rate_A_B)
61.
62.     # Calculamos pobreza con líneas de A
63.     baseEC <- pobrezaaing(baseA_EC)
64.
65.     # Nuevo objeto de diseño
66.     svyEC <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = baseEC)
67.
68.     EC <- as.numeric(svymean(~plp+plp_e, design = svyEC))
69.     G <- (EC - ObsA_SPS)
70.
71.     ##### EFECTO REDISTRIBUCIÓN #####
72.     # Deflactamos a pesos de A y aplicamos la tasa de crecimiento de B/A
73.     baseB_ERD <- mutate(B_SK, ictpc = (ictpc/defEC)*rate_B_A)
74.
75.     baseB_ERD <- pobrezaaing(baseB_ERD)
76.
77.     svy_ERD <- svydesign(ids = ~upm , weights = ~factor,
strata=~est_dis, data = baseB_ERD)
78.
79.     ERD <- as.numeric(svymean(~plp+plp_e, design = svy_ERD))
80.     RD <- (ERD - ObsA_SPS)
81.

```

```

82. ##### EFECTO INFLACIÓN #####
83.
84. baseA_EI <- mutate(A_SK, ictpc = ictpc*def)
85.
86. baseA_EI <- mutate(baseA_EI,
87.   # Se identifica a los hogares bajo lp1
88.   plp_e=case_when(ictpc < Ilp1_urb & rururb==0 ~ 1,
89.                   ictpc>=Ilp1_urb & rururb==0 & !is.na(ictpc) ~ 0,
90.                   ictpc < Ilp1_rur & rururb==1 ~ 1,
91.                   ictpc>=Ilp1_rur & rururb==1 & !is.na(ictpc) ~ 0),
92.   # Se identifica a los hogares bajo lp2
93.   plp =case_when(ictpc < Ilp2_urb & rururb==0 ~ 1,
94.                  ictpc>=Ilp2_urb & rururb==0 & !is.na(ictpc) ~ 0,
95.                  ictpc < Ilp2_rur & rururb==1 ~ 1,
96.                  ictpc>=Ilp2_rur & rururb==1 & !is.na(ictpc) ~ 0))
97.
98. svy_EI <- svydesign(ids = ~upm , weights = ~factor, strata=~est_dis, data = baseA_EI)
99.
100. EI <- as.numeric(svymean(~plp+plp_e, design = svy_EI))
101. I <- (EI - ObsA_SPS)
102.
103. # Generamos la de resultados y el residual
104. results <- round(data.frame("A" = ObsA, "B" = ObsB, "Cambio" = dif, "Crecimiento"
= G, "Redistribución" = RD, "Inflación" = I, "Programas Sociales" = Dif_PS)*100,1)
105. results <- mutate(results,
  Residual = Cambio - (Crecimiento + Redistribución + Inflación + Programas.Sociales))
106. rownames(results) <- c("Pobreza por ingresos", "Pobreza extrema por ingresos")
107. results
108. }

```

Ejemplo de la salida en la consola de R:

```

> source("DR_descomp_SK.R")
> DR_descompSK(baseA, baseB, baseA_SPS, baseB_SPS, def = 1.231968714, lp1_urb = 1544.07,
+   lp1_rur = 1164.75, lp2_urb = 3325.40, lp2_rur = 2316.57, Ilp1_urb = 2086.21,
+   Ilp1_rur = 1600.18, Ilp2_urb = 4158.35, Ilp2_rur = 2970.76)

```

	A	B	Cambio	Crecimiento	Redistribución	Inflación	Programas.Sociales	Residual
Pobreza por ingresos	49.9	43.5	-6.4	-3.5	-2.5	1.1	-1.5	0.0
Pobreza extrema por ingresos	14.0	12.1	-1.9	-1.6	-2.2	3.1	-0.7	-0.5